

Consumer's Financial Habits on Server-Based Electronic Money as It Affects Their Financial Behavior: Moderated By Monthly Transactions

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Abstract

This study investigates the impact of various financial habits—specifically saving, spending, donating, investing, and credit/loan behaviors—on consumer financial behavior, with transaction frequency serving as a moderating factor. Grounded in the Theory of Planned Behavior, the research utilized a quantitative approach, collecting data from a targeted group of electronic money users (GCash) through purposive sampling. The multiple regression analyses revealed that all financial habits, except for credit/loan behaviors, significantly influence financial behavior. Additionally, moderation analysis demonstrated that higher transaction volumes enhance the relationship between spending habits and financial behavior, emphasizing the role of mobile wallet usage in reinforcing these spending patterns. These findings provide valuable insights for fintech companies, enabling them to improve platform usability and financial management tools, ultimately promoting responsible financial habits among users. Consumers can gain a clearer understanding of how mobile wallets affect their financial decisions, encouraging more informed spending and saving practices. Future research should delve deeper into long-term behavioral trends, qualitative insights, and the psychological factors driving digital financial behavior, offering a broader perspective on how technology shapes financial habits over time. By exploring these areas, the evolving landscape of consumer finance in the digital age can be understood further.

CCS Concepts

• **Applied computing** → Law, social and behavioral sciences; Economics.

Keywords

Server-Based Electronic Money, GCash, style, Financial Behavior, insert

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1 INTRODUCTION

In recent years, rapid technological innovations have significantly transformed consumer behavior, particularly through the rising popularity of e-commerce and the use of electronic wallets (e-wallets) for online transactions. As consumers seek more convenient and efficient ways to shop, traditional payment methods—such as cash and credit cards—are increasingly being replaced by more practical and secure alternatives [1].

Advancements in digital wallet technology have played a crucial role in this transition. E-wallets allow users to store payment information and make transactions quickly and securely from their mobile devices or computers. Features such as biometric security, real-time transaction notifications, and digital receipt tracking enhance the user experience and build trust among consumers. The shift toward e-commerce and e-wallets has been accelerated by the global pandemic, which forced many businesses to adapt to online selling and consumers to seek contactless payment options. This trend highlights the importance of digital wallets as not only a safer alternative but also a convenient choice for everyday transactions [2].

Given these changes, it is essential to explore the implications of digital wallets on consumer behavior. Understanding how these technologies affect purchasing decisions, trust in online platforms, and overall satisfaction can provide valuable insights for businesses aiming to meet the evolving needs of their customers in this dynamic landscape.

Server-based electronic money systems offer enhanced security features designed to mitigate the risks associated with electronic transactions. An e-wallet, as defined by the Bank for International Settlements, is a non-cash payment device that incorporates stored value or prepaid products on an electronic device utilized by the consumer. Supporting this concept, a study conducted by Lai [3] indicated that increased security measures are positively correlated with a heightened intention to utilize digital payment technologies. Concerns surrounding the risks of theft and loss associated with carrying cash and cards render security a critical factor influencing consumer decision-making in relation to electronic payments.

This trend is particularly pronounced in developing countries such as the Philippines, where the demand for cashless transactions has accelerated following the endorsement of e-banking and e-wallets by the Bangko Sentral ng Pilipinas, the country's banking regulatory authority [4]. As of 2023, the Philippines has witnessed a considerable surge in the adoption of e-wallets, exemplified by

GCash, which has attained a notable milestone of over 60 million registered users nationwide [5]. This widespread acceptance underscores GCash's significance as a virtual wallet that facilitates secure, rapid, and convenient money transfers, thereby transforming the digital payments landscape in the Philippines.

Initially launched in 2004, GCash gained substantial traction during the COVID-19 pandemic, buoyed by regulatory support and strategic innovations from telecommunications leaders Smart Communications and Globe Telecom. Presently, GCash serves a crucial role in promoting financial inclusion, offering a versatile platform for various transactions, including bill payments, money transfers, and mobile commerce [5].

This study aimed to investigate the effects of server-based electronic money, specifically focusing on GCash, on consumers' financial behaviors. It explored various aspects such as saving, spending, donating, investing, and credit or loan habits to understand how factors like specific financial practices and the accessibility of GCash influenced consumer use of the platform. This research sought to illuminate how these factors shaped financial decision-making processes. Additionally, the study examined the relationship between financial habits and the average frequency of transactions per month, analyzing how this usage frequency affected consumers' overall financial behavior. By establishing this connection, the research aimed to provide insights into the impact of GCash on the financial choices and practices of its users.

2 LITERATURE REVIEW

2.1 Server-Based Electronic Money and Financial Behavior

The emergence of server-based electronic money (SBEM), including digital wallets and mobile payment platforms, has transformed consumer financial behavior by enhancing convenience, accessibility, and transaction speed. Operating through centralized servers, SBEM enables secure, real-time transactions without the need for physical cash or traditional banking. Studies show that FinTech innovations have led to more frequent small-value transactions, increased reliance on mobile devices, and a shift toward cashless economies, reflecting both technological and psychological changes. SBEM also promotes financial inclusion and literacy, especially among underserved populations, by simplifying saving, borrowing, and risk management. However, without proper financial education, these tools may also encourage impulsive spending, highlighting the need for a balanced understanding of their benefits and risks. [6] [7]

2.2 Financial Habits and Their Impact on Overall Financial Behavior

Understanding consumers' financial habits, such as donating, investing, saving, and borrowing, provides crucial insights into their overall financial behavior, as these actions are interconnected within a broader financial planning framework. Influenced by factors like financial satisfaction, social norms, and financial literacy, these habits reflect varying levels of financial resilience and planning. For instance, consistent saving and investing often signal proactive financial management, while responsible borrowing and

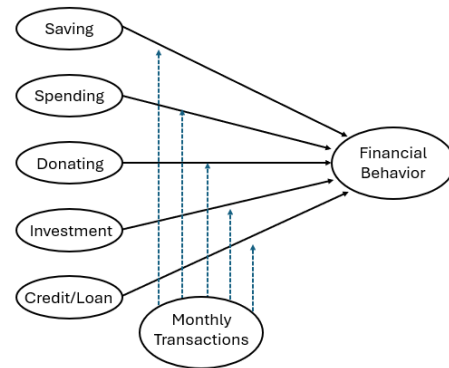


Figure 1: Conceptual Framework

charitable giving may indicate confidence and discipline. However, many consumers lack sufficient financial knowledge, which can lead to poor decisions and financial instability. Studies emphasize that financial education and behavioral insights are essential for promoting informed, responsible financial behavior and long-term well-being. [8] [9]

2.3 Monthly Transactions as they affect Financial Habits and Behavior

The frequency of monthly transactions on server-based electronic money (SBEM) platforms significantly shapes users' financial habits by promoting convenience and streamlining decision-making. As consumers increasingly rely on digital wallets and mobile payments, they adopt new spending behaviors characterized by frequent, smaller transactions. This shift, driven by ease of access and speed, can lead to improved budgeting through transaction tracking but also raises concerns about impulsive spending. Regular use of SBEM fosters deeper engagement with digital finance, encouraging users to explore advanced features like automated savings and embedded credit tools. Aligned with the Technology Acceptance Model, this trend highlights how perceived usefulness and ease of use drive adoption, ultimately transforming traditional financial practices into more dynamic, data-driven approaches. [10]

3 METHODOLOGY

3.1 Research Design

This study used a quantitative approach to explore how financial habits—such as saving, spending, donating, investing, and managing credit or loans—affect consumer financial behavior in the context of server-based electronic money (SBEM). It also examined whether the average number of monthly transactions moderates these relationships, revealing how transaction frequency can influence the strength of financial habit effects. Guided by the Theory of Planned Behavior (TPB), the research identified key psychological and behavioral factors shaping financial actions in a digital environment. Figure 1 shows how TPB concepts were integrated to provide a comprehensive analysis of how individual financial habits collectively influence broader financial behaviors, offering

insights into the evolving nature of consumer engagement with digital financial platforms. [11]

3.2 Sampling

The researchers employed a purposive sampling technique to select 300 participants for their study, ensuring demographic representation across age, gender, income/allowance, and occupation. Participants were comprehensively briefed on the study's objectives and guaranteed confidentiality in accordance with ethical standards. This sampling method was deemed suitable as it targeted individuals capable of providing valuable insights into how various financial behaviors, namely saving, spending, donating, investing, and managing credit or loans, impact engagement with server-based electronic money platforms. Furthermore, the study aims to investigate how the average number of monthly transactions influences these relationships. The selection criteria encompassed being a Filipino user of GCash, aged between 18 and 60, and actively participating in at least one of the specified financial behaviors.

3.3 Research Instrument

The study utilized online questionnaires distributed through Google Forms and social media to collect primary data on respondents' financial behaviors related to various features of GCash. A 5-point Likert Scale was used in the questionnaire to evaluate attitudes and perceptions across several sections, including demographic profiling, saving behaviors influenced by GCash's GSave, spending habits shaped by cashless transactions, and the impact of GCash's donation platform. Additional sections examined the effects of GInvest on investment behavior, GCash's credit features, and average monthly transaction patterns among users. Ultimately, the research aimed to quantify how digital payment tools like GCash influence consumer financial decision-making and encourage responsible financial behavior.

3.4 Statistical Treatment

In this study, multiple regression analysis was employed to investigate the extent to which specific financial habits—namely saving, spending, donating, investing, and managing credit or loans—predict consumer financial behavior in the context of server-based electronic money. This statistical method allowed for the simultaneous examination of multiple independent variables to determine their individual and collective influence on the dependent variable, financial behavior [12]. To further explore the dynamics of this relationship, a moderation analysis was conducted to assess whether the frequency of monthly transactions moderated the effect of financial habits on financial behavior [13]. By incorporating this interaction term, the analysis aimed to reveal whether the strength or direction of the relationship between financial habits and behavior varied depending on the level of monthly engagement with server-based electronic money platforms. This dual approach provided a more nuanced understanding of how digital financial activity shapes consumer behavior patterns.

4 Results and Discussions

4.1 Assumption Check

Before conducting multiple regression analyses, the dataset underwent diagnostic tests to ensure model reliability, including checks for autocorrelation, multicollinearity, and residual normality. The regression model, grounded in the study's conceptual framework, examined how five independent variables—Saving Habit, Spending Habit, Donating Habit, Investment Habit, and Credit/Loan Habit—affect Consumer Financial Behavior. Collinearity statistics showed acceptable VIF values (1.26–1.80), indicating no serious multicollinearity. The Durbin-Watson statistic of 1.73 suggested no significant autocorrelation, and the Shapiro-Wilk test confirmed that residuals were approximately normally distributed. These results validate the model's assumptions, supporting the reliability and validity of its findings. [14] [15]

4.2 Model Fit Measures

The comprehensive evaluation of model fit indicators show that the regression model is both statistically significant and reliable. The correlation coefficient ($R = 0.633$) reveals a strong positive relationship between the independent variables and Consumer Financial Behavior, while the coefficient of determination ($R^2 = 0.400$) indicates that 40% of the variance in the dependent variable is explained by the model. The RMSE of 0.502 suggests that prediction errors are minimal, reinforcing the model's accuracy. Additionally, the F-test result ($F(5, 927) = 39.6, p < 0.001$) confirms the overall significance of the model, meaning at least one independent variable has a meaningful impact. Together, these metrics demonstrate that the model provides a solid and dependable fit for the data. [16]

4.3 The Financial Habits of Consumers and Their Impact on Financial Behavior

This study aimed to investigate how specific financial habits, such as budgeting, saving, spending, donating, and managing credit, affect consumer behavior on GCash, a widely used mobile wallet in the Philippines. Using multiple regression analysis, the research examined how these habits influence users' interactions with the platform, including transaction frequency, fund management, and usage for bill payments and online shopping. The study also explored the moderating role of monthly transaction frequency, offering insights into how consistent engagement with GCash can shape financial decision-making and promote better financial management strategies.

The results in Table 1 revealed that four out of five financial habits, investment, saving, donating, and spending, significantly influence consumer financial behavior on GCash. Investment habits had the strongest impact, followed by spending and saving habits, all of which were positively associated with effective financial management and platform engagement. Donating habits also showed a meaningful influence, suggesting that users value GCash's convenience for charitable contributions. The low standard errors in the regression estimates indicate high precision and reliability of the findings. Overall, the study highlights the critical role of digital platforms like GCash in fostering positive financial habits and

Table 1: Coefficients for the Regression Model

Hypothesis	Predictor	β	SE	t	p	Interpretation
H1	SH	0.178*	0.047	3.820	< .001	Significant
H2	SPH	0.115*	0.038	3.047	0.003	Significant
H3	DH	0.144*	0.048	3.014	0.003	Significant
H4	IH	0.243*	0.050	4.896	< .001	Significant
H5	CLH	-0.003	0.047	-0.056	0.955	Not Significant

Dependent Variable: Consumer Financial Behavior; β – Coefficients; *Highly significant at 0.01

suggests that enhancing user experience could further improve financial engagement and behavior. [17]

4.4 Moderation Analysis

This study investigated how average monthly transactions on GCash moderate the relationship between users' financial habits and their overall financial behavior. Using multiple regression analysis with an interaction term, the research found that only one moderation effect, between spending habit and transaction frequency, was statistically significant ($t = -4.6$, $\beta = -0.1629$, $p < 0.001$). This result indicates that frequent GCash transactions significantly influence how spending habits affect financial behavior, suggesting that users who transact more often may experience different behavioral outcomes compared to less frequent users. The findings highlight the importance of usage patterns in shaping financial behaviors on digital platforms. [18]

The simple slope analysis indicates that at low and average levels of monthly transactions, the effect of spending habits on financial behavior is positive but relatively modest. These results suggest that as monthly transactions increase, the positive impact of spending habits on financial behavior becomes more pronounced. Therefore, users with low to average monthly transactions in GCash may utilize the platform to manage and control their financial transactions effectively, while also spending more when making purchases through GCash.

5 CONCLUSION AND RECOMMENDATIONS

This study offers a detailed analysis of how consumer financial habits—specifically saving, spending, donating, and investing—affect financial behavior on the GCash platform, with a focus on the moderating role of monthly transaction frequency. The findings reveal that each habit significantly influences user engagement and financial outcomes, with investment and spending habits showing particularly strong effects. Notably, the frequency of monthly transactions intensifies the impact of spending habits, suggesting that users who transact more often experience greater behavioral shifts. These insights emphasize the importance of fostering positive financial habits and optimizing digital platform features to enhance financial management. The study also recommends improving user interfaces and developing targeted strategies to guide spending

behavior, ultimately supporting responsible financial practices and contributing to a more sustainable digital economy.

References

- [1] Hidayati, Istiqlaliah & Polytechnique, Telkom. (2012). Evaluating the role of L1 in teaching receptive skills and grammar in EFL classes. *Indonesian Journal of Applied Linguistics*. 1. 10.17509/ijal.v1i2.82.
- [2] Jílková, P., & Králová, P. (2021). Digital consumer behaviour and eCommerce trends during the COVID-19 crisis. *International Advances in Economic Research*, 27(1), 83–85. <https://doi.org/10.1007/s11294-021-09817-4>.
- [3] Lai, P. C. (2016). Design and security impact on consumers' intention to use single platform e-payment. *Interdisciplinary Information Sciences*, 22(1), 111–122. <https://doi.org/10.4036/iis.2016.R.05>.
- [4] Bangko Sentral ng Pilipinas. (2020). Digital payments transformation roadmap2020–2023. https://www.bsp.gov.ph/Media_And_Research/Primers%20Faq/Digital%20Payments%20Transformation%20Roadmap%20Report.pdf
- [5] Globe Telecom. (2022, May 24). GCash achieves new milestone with over 60M registered users. <https://www.globe.com.ph/about-us/newsroom/consumer/gcash-new-milestone-over-60m-registered-users>.
- [6] Gomber, P., Koch, J.-A., & Siering, M. (2017). Digital Finance and FinTech: current research and future research directions. *Journal of Business Economics*.
- [7] Tiony, O. K., & Yin, Y. K. (2023). The Impact of Digital Financial Services on Financial Inclusion in Kenya. *American Journal of Industrial and Business Management*, 13, 593–628. doi: 10.4236/ajibm.2023.136035.
- [8] Yeo, K. H. K., Lim, W. M., & Yui, K. J. (2024). Financial planning behaviour: a systematic literature review and new theory development. *Journal of Financial Services Marketing*.
- [9] Raaij, W. F. (2016). *Understanding Consumer Financial Behavior: Money Management in an Age of Financial Illiteracy*. Springer.
- [10] Ma, Q., & Liu, L. (2004). The Technology Acceptance Model: A Meta-Analysis of Empirical Findings. *Journal of Organizational and End User Computing*, 16(1), 59–72. Retrieved from ResearchGate
- [11] Bosnjak, M., Ajzen, I., & Schmidt, P. (2020). The Theory of Planned Behavior: Selected Recent Advances and Applications. *Europe's Journal of Psychology*, 16(3), 352–356. <https://doi.org/10.5964/ejop.v16i3.3107>
- [12] Nayeib, H. (2020). Multiple Regression Analysis. In *Advanced Statistics for Testing Assumed Causal Relationships* (pp. 1–46). Springer. https://doi.org/10.1007/978-3-030-54754-7_1
- [13] Memon, M. A., Cheah, J.-H., Ramayah, T., Ting, H., Chuah, F., & Cham, T. H. (2019). Moderation Analysis: Issues and Guidelines. *Journal of Applied Structural Equation Modeling*, 3(1), i–xi.
- [14] Sevier, F. A. C. (1957). Testing the assumptions underlying multiple regression. *Journal of Experimental Education*, 25, 323–330. <https://doi.org/10.1080/00220973.1957.11010578>
- [15] O'Brien, R. M. (2007). A caution regarding rules of thumb for variance inflation factors. *Quality & Quantity*, 41(5), 673–690. <https://doi.org/10.1007/s11135-006-9018-6>
- [16] Miles, J., & Shevlin, M. (2001). *Applying regression and correlation: A guide for students and researchers*. SAGE Publications. <https://doi.org/10.4135/9781849208963>
- [17] Kutner, M. H., Nachtsheim, C. J., & Neter, J. (2004). *Applied Linear Regression Models* (4th ed.). McGraw-Hill/Irwin.
- [18] McLeod, S. (2025). *Moderating Variable in Statistics*. Simply Psychology.